SEC:: SR. JEE MAINS APEX | GTM-13 | Date: 07.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
A	C	D	B	C	B	C	C	A	A
11	12	13	14	15	16	17	18	19	20
B	D	D	A	A	A	A	A	D	C
21	22	23	24	25	26				
748	15	16	3	7					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	C	A	D	C	D	B	B	C	A
36	37	38	39	40	41	42	43	44	45
D	B	C	B	C	A	C	A	C	C
46	47	48	49	50					
197	5	1	5	55					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	D	A	A	A	C	C	C	A	B
61	62	63	64	65	66	67	68	69	70
C	C	A	A	D	D	C	A	B	C
71	72	73	74	75					
17	103	5	55	96					

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PHYSICS

1. $K.E = \frac{P^2}{2m}$

Θ two particles are having same magnitude of momentum,

$\therefore \frac{K_B}{K_A} = \frac{m_A}{m_B} = \frac{V_B}{V_A}$

2. $\frac{L}{3L} = \frac{m_1 r_1^2 w_1}{m_2 r_2^2 w_2}$

$\frac{1}{3} = \frac{m_1}{m_2}\frac{r_1^2}{r_2^2}\frac{T_2}{T_1}$

As $T^2 \propto r^3$

$\frac{1}{3} = \frac{m_1}{m_2}\frac{T_1^{4/3}}{T_2^{4/3}}\frac{T_2}{T_1}$

$\frac{T_1}{T_2} = \frac{1}{27}\left(\frac{m_2}{m_1}\right)^3$

3. For mono atomic, $C_V = \frac{3}{2}R$

Diatomic, $C_V = \frac{5}{2}R$

$\therefore \text{ratio} = \frac{3}{5}$

4. $F = m\frac{\Delta V}{\Delta t}$

$= (150gm)\frac{(20m/s - 0)}{0.1}$

$= 30\ \text{N}$

5. At resonance,

$i_o = \frac{\xi_o}{R}$

$\therefore i_o^1 = \frac{\xi_o}{R/2}$

$= 2i_o$

6. $F = \frac{I}{C}A$

$2.4\times10^{-4} N = \frac{360\times10^{+4} W/m^2}{3\times10^8 m/s}\cdot A$

$\therefore A = 0.02m^2$

7. $B.E = (\Delta m)C^2$

$\Delta m = \frac{B.E}{C^2} = \frac{18\times10^8 J}{9\times10^{16} m^2/\text{sec}^2} = 20\mu g$

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8. $Y=\left(A+\overline{B}\right)\cdot\left(B\cdot\overline{A}\right)$

$=A\cdot\left(B\cdot\overline{A}\right)+\overline{B}\cdot\left(B\cdot\overline{A}\right)$

$=0$

9. If T_1 is temperature in isothermal expansion $a \to b$ and T_2 is temperature in isothermal compression $c \to d$, then for $b \to c$

$T_1V_b^{\gamma-1}=T_2V_c^{\gamma-1}$

for $d \to a$

$T_2V_d^{\gamma-1}=T_1V_a^{\gamma-1}$

$\therefore \frac{V_a}{V_d}=\frac{V_b}{V_c}$

10. Conceptual

11. $i=\frac{\xi}{R_{eq}+r}=1amp$

$V=\xi-ir$

$=3(1)\ (1)$

$=2\text{V}$

12. $P=\sqrt{2m(K.E)}$

$\therefore P_A:P_B:P_C=\sqrt{m_A}:\sqrt{m_B}:\sqrt{m_C}$

$=\sqrt{0.4}:\sqrt{1.2}:\sqrt{1.6}$

$=1:\sqrt{3}:2$

13. $\sin c=\frac{\mu_r}{\mu_d}$

$\sin 45^\circ=\frac{\mu_2}{\mu_1}$

$\frac{\mu_1}{\mu_2}=\sqrt{2}$

14. $Y=49000\frac{m}{l}$

$\frac{\Delta Y}{Y}=\frac{\Delta m}{m}+\frac{\Delta l}{l}$

$\therefore \frac{\Delta Y}{Y}\times 100=\frac{\Delta m}{m}\times 100+\frac{\Delta l}{l}\times 100$

$=\frac{5}{500}\times 100+\frac{0.02}{2}\times 100$

$=1\ \%+1\ \%$

$=2\%$

15. Conceptual

16. $\lambda=\frac{h}{P}=\frac{h}{\sqrt{2mk}}$

As de-broglie wavelength is same, momentum must be equal

$\therefore \sqrt{2m_ek_e}=\sqrt{2m_pk_p}$

$\frac{k_p}{k_e}=\frac{m_e}{m_p}=\frac{1}{1836}$

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17. $P+\rho gh+\frac{1}{2}\rho v^2=\text{Constant}$

18. $x=d_s-d_m$

$=(30)2\pi r_s-\pi r_m$

$=\pi\left[60r_s-r_m\right]$

$=139.4\text{ m}$

19. d = 2cm + (0.01) × 2

= 2.02 cm

$V=\frac{\pi d^3}{6}$

$\therefore density=\frac{mass}{vol}=\frac{8.635gm}{\frac{\pi}{6}(2.02)^3\,cm^3}$

$2.0\,gm/cm^3$

20. $V=\frac{kq}{r}$

$\frac{q_A}{q_B}=\frac{a}{b}$

21. $\Delta R = R_0\alpha\Delta T$

$\therefore 0.2\Omega = 10\alpha(100)...(1)$

$0.95\Omega = 10\alpha t...(2)$

t = 47.5° C

t = 748 k

22. The metal plate can be considered as five equal parts each of mass 2kg and dimensions (1cm × 1cm)

X_{CM} = 1.5cm

$Y_{CM}=\frac{6\times(0.5)+2\times(1.5)+2\times(1.5)}{10kg}$

$=0.9$

$\frac{X_{CM}}{Y_{CM}}=\frac{1.5}{0.9}=\frac{15}{9}$

23. $\frac{(15)\frac{V}{4L}}{8\cdot\frac{V}{2L}}=\frac{15}{16}$

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24. W = (Fl)N

= (ilBl)N

$= \frac{\Delta\phi}{\Delta t \cdot R} Bl^2$

$= \frac{N^2 Bl^2}{(\Delta t) \cdot R} Bl^2$

$= \frac{N^2 B^2 l^4}{(\Delta t) R}$

$= 3.24 \times 10^{-6} J$

25. $heg = \frac{4T}{R}$

$R = \frac{4T}{heg} \Rightarrow 2R = \frac{8T}{heg}$

$= 7 \times 10^{-2} m = 7cm$

CHEMISTRY

26. $CH_3 - \underset{\underset{H}{|}}{C}(CH_3) - \underset{\underset{H}{|}}{C}H - \underset{\underset{H}{|}}{C}(CH_3) - CH_3$

27. Borax bead test ——— $MSO_4 \xrightarrow[\Delta]{Na_2B_4O_7} M(BO_2)_2 \rightarrow MBO_2 \rightarrow M$

Charcoal cavity test ——— $MSO_4 \xrightarrow[\Delta]{Na_2CO_3} MCO_3 \rightarrow MO \rightarrow M$

Cobalt nitrate test ——— $MCO_3 \rightarrow MO \xrightarrow[+\Delta]{CO(NO_3)_2} CoOMO$

Flame test ——— $MCO_3 \rightarrow MCl_2 \rightarrow M^{2+}$

28. Cl_2, Br_2, I_2 Can undengo disproportion Reaction

29. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

Given Mass of glucose = 900 gram

Molar mass of glucose = 180 gm/mole

So, moles of glucose $\Rightarrow \frac{900}{180} = 5$

1 mole of glucose require 6 moles of oxygen gas

So, 5 moles of glucose will require 30 moles of oxygen gas

Moles of oxygen gas = 30

Molar mass of oxygen gas $\Rightarrow$ 32 g/mol

So, mass of oxygen gas required $\Rightarrow$ 30 × 32

$\Rightarrow$ 960 gram.

30. $Fe_4[Fe(CN)_6]_3 xH_2O$ ——— Prussian Blue

$[Fe(CN)_5 NOS]^{4-}$ ——— Violet

$[Fe(SCN)]^{2+}$ ——— Blood Red

$(NH_4)_3 PO_4 12MoO_3$ ——— Yellow

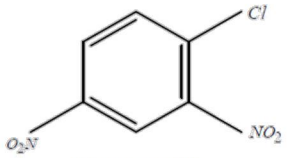
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31. Cl, S ——— Electrons with highest negative electron gain entalpy

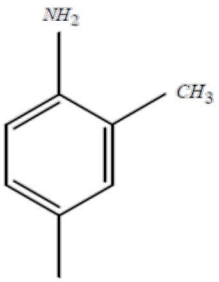
Ge, As ——— Elements which show properties of both metals and non-metal

Fr, Ra ——— Elements with larger atomic sixe

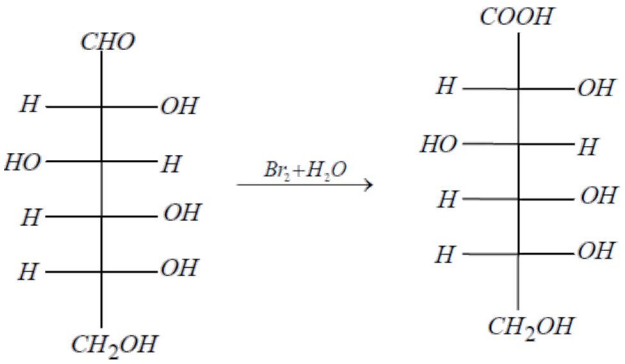
F, O ——— Elements with highest electronegativity

32. 

CompoundA 2-chloro-1,5- dinitrobenzene



CompoundB is 4-ethyl-2-methylaniline

33. 

34. $2Fe^{+3} + 2I^{-1} \rightarrow 2Fe^{+2} + I_2$

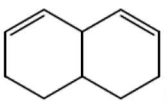
$2Fe^{+2} + S_2O_8^{-2} \rightarrow 2Fe^{+3} + 2SO_4^{-2}$

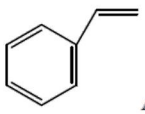
35. SN^2 Reactivity: $CH_3 > 1RX > 2° RX > 3° RX$

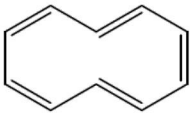
36. Statement I is correct but Statement II is Incorrect

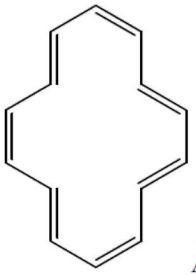
$N(CH_3)_3$ and $P(CH_3)_3$ both are Lewis base and acts as ligand, However, $P(CH_3)_3$ has a π-acceptor character.

37. Due to inert pair effect +1 oxi state stability increase from top to bottom in IIIA group

38.  non ARO

 Aromatic

 non Aromatic

 Aromatic

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39. $X \rightleftharpoons Y$

$Y \rightleftharpoons Z$

$Z \rightleftharpoons \omega$

$X \rightleftharpoons \omega$

$k_1 = 1$

$k_2 = 2$

$k_3 = 4$

$k_1 . k_2 . k_3$

$k = 1 \times 2 \times 4$

$k = 8$

40. NH_3 ——— Trigonal pyramidal

BrF_5 ——— Square pyramid

PCl_5 ——— Trigonal bipyramidal

CH_4 ——— Tetrahedal

41. $[Fe(H_2O)_6]^{+2} \rightarrow d^6 WFL \rightarrow t_{2g}^4 eg^2$ (even no of e^- in t_{2g})

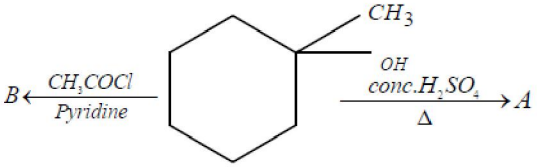
$[Co(H_2O)_6]^{+2}$ $d^7 \rightarrow WFL \rightarrow t_{2g}^5 eg^2$ (odd no of e^- in t_{2g})

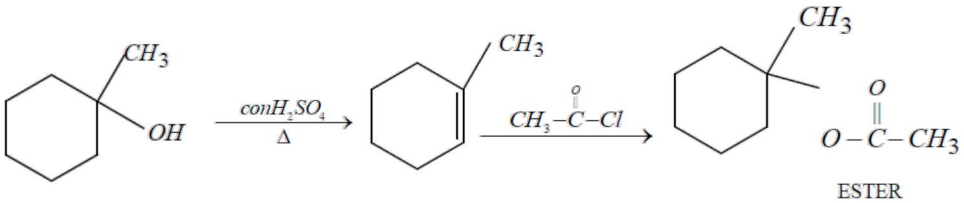
$[Co(H_2O)_6]^{+3} \rightarrow d^6 SFL \rightarrow t_{2g}^6 eg^0$ (even no of e^- in t_{2g})

$[Cu(H_2O)_6]^{+2}$ $d^4 \rightarrow$ WFL $t_{2g}^6 eg^3$ (even no e e^- in t_{2g})

$[Cr(H_2O)_6]^{+2}$ $d^4 \rightarrow$ WFL $t_{2g}^3 eg^1$ (odd no of e^- in t_{2g})

Thus three complex show even no of e^- in t_{2g} orbital.

42. 

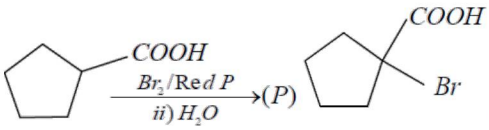


ESTER

43. $CoCl_3 . xNH_3 + AgNO_3 \rightarrow 2 mole of\ AgCl$

So complex is $\left[Co(NH_3)_5 Cl\right]Cl_2$

$x = 5$ and ox No. of Co is +3

44. 

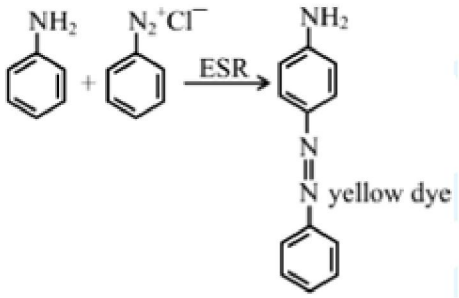
HVZ Reaction

45. $2S_2^{+2}O_3^{2-} + I_2 \rightarrow S_4^{2.5}O_6^{2-} + 2I^-$

$S_2^{+2}O_3^{2-} + 5Br_2 + 5H_2O \rightarrow 2S^{+6}O_4^{2-} + 4Br^- + 10H^+$

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46.



The limiting reagent is benzenediazonium chloride,
Hence 1 mole of product is formed.
It's weight is 197 g

47. $\Delta T_f = i.k_b m$

$$(100.52-100) = i \times 0.52 \times \left(\frac{10}{200} \times \frac{1000}{100}\right)$$

$$= 0.52 = \frac{i \times 0.52}{2}$$

$$i = 2$$

$$1+(n-1)\alpha = 2$$

$$1+(3-1)\alpha = 2$$

$$2\alpha = 1$$

$$\alpha = 0.5 = 5 \times 10^{-1}$$

48. 1st order w.r.t A & zero order w.r.t B

49. 1° amine will give this test

50. $\omega = -nRT\,ln\left(\frac{V_2}{V_1}\right)$

$$= -1 \times 0.08206 \times 291.15\,ln\left(\frac{100}{10}\right)$$

= -55.0128

Work done by system ≈ 55 atm lit.

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51. $\frac{2(8)+\alpha}{3}=6\Rightarrow\alpha=2$

$\frac{2\left(\frac{15}{2}\right)+\beta}{3}=6\Rightarrow\beta=3$

$r_1=\sqrt{(6-2)^2+(6-3)^2}=5\,;\qquad r_2=\sqrt{(6-8)^2+\left(6-\frac{15}{2}\right)^2}$

$\therefore\alpha+\beta+4\left(r_1^2+r_2^2\right)\qquad=\frac{5}{2}$

$=2+3+4\left(25+\frac{25}{4}\right)=130$

52. $I(x)=\int\frac{6}{\sin^2 x(1-\cot x)^2}\,dx=\int\frac{6}{(\sin x-\cos x)^2}\,dx$

$=\int\frac{6\sec^2 x}{(\tan x-1)^2}\,dx\quad put\quad \tan x-1=t$

$such\ dx=dt$

$I(x)=\ {}^{6}\int\frac{1}{t^2}dt\quad=\frac{-6}{t}+C=\frac{-6}{\tan-1}+C$

$I(0)=3\qquad\Rightarrow 3=6+C\qquad\Rightarrow C=-3$

$\therefore I(x)=\ \frac{6}{1-\tan x}-3$

$I\left(\frac{\Pi}{12}\right)=\frac{6}{1-\left(2\sqrt{3}\right)}-3\ \ \frac{6}{\sqrt{3}-1}-3$

$=\frac{6\left(\sqrt{3}+1\right)}{2}-3\quad=3\sqrt{3}$

53 $f(x)=(x-2)^{2/3}(2x+1)$

$f'(x)=(x-2)^{2/3}(2)+\frac{2}{3}(x-2)^{1/3}(2x+1)$

$f'(x)=\ 2(x-2)^{2/3}+\frac{2(2x+1)}{(x-2)^{1/3}}$

$\Rightarrow f'(x)=\ \frac{2(x-2)+2(2x+1)}{(x-2)^{1/3}}=\frac{6x-2}{(x-2)^{1/3}}$

For critical points be have $f'(x)=0$ or $f'(x)$ does not exist so $x=\frac{1}{3};2$ are the critical points

54. $(a,b)=\{(2,8),(7,7),(12,6),(17,5),(22,4),(27,3),(32,2),(37,1)\}$

$\therefore m=8\sum_{n=1}^{m}\left(1-i^n\right)=\ x+y$

$\Rightarrow x+y=\ (1-i)+2+2\quad=5-i$

$m+x+y=\ 8+5-1=\ 12$

55. Given lines are

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$$L_1 : \bar{r} = 2\hat{i} + \hat{j} + 3\hat{k} + \lambda\left(\hat{i} - 3\hat{j} + 4\hat{k}\right)$$

$$L_2 : \bar{r} = 2\hat{i} + 3\hat{j} + 5\hat{k} + \mu\left(2\hat{i} + 3\hat{j} + \hat{k}\right)$$

Shortest distance between the lines

$$= \frac{\begin{Vmatrix} 0 & 1-3 & 3-5 \\ 1 & -3 & 4 \\ 2 & 3 & 1 \end{Vmatrix}}{\left|\left(\hat{i} - 3\hat{j} + 4\hat{k}\right) \times \left(2\hat{i} + 3\hat{j} + \hat{k}\right)\right|} = \frac{32}{\sqrt{355}}$$

56. Prime factors of 2310 *are* 2,3,5,7,11

$\therefore A = \{2,3,5,7,11\}$

$$f(x)\left[\log_2\left(x^2 + \left[\frac{x^3}{5}\right]\right)\right]$$

Range of $f = \{2,3,5,6,8\}$

No of 1 – 1 function = 120

57. $$\int_o^a f(x)\,dx = e^{-a} + 4a^2 + a - 1$$

Differentiating w.r.t +a

We have $f(a) = -e^{-a} + 8a + 1$

$$\therefore f(x) = -e^{-x} + 8x + 1$$

$$y = C_1\left(-e^{-x} + 8x + 1\right) + C_2$$

Differentiating w.r.t +x

$$\frac{dy}{dx} = C_1\left(e^{-x} + 8\right)$$

$$\frac{d^2y}{dx^2} = -C_1 e^{-x}$$

$$\Rightarrow e^x \frac{d^2y}{dx^2} = -C_1$$

$$\Rightarrow e^x \frac{d^2y}{dx^2} = -\left(\frac{\frac{dy}{dx}}{8 + e^{-x}}\right)$$

$$\Rightarrow e^x\left(8 + e^{-x}\right)\frac{d^2y}{dx^2} = -\frac{dy}{dx}$$

$$\Rightarrow \left(8e^x + 1\right)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$$

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58. $f(x)=4\cos^3 x+3\sqrt{3}\cos^2 x-10$

$f'(x)=\left(12\cos^2 x+6\sqrt{3}\cos x\right)(-\sin x)$

$\Rightarrow f'(x)=-6\sin x\ \cos x\left(2\cos x+\sqrt{3}\right)$

$f'(x)=0\Rightarrow \sin x=0\ \cos x=0\ 8\cos x=\dfrac{-\sqrt{3}}{2}$

$f'(x)=-12\ \sin x\ \cos x\left(\cos x-\left(\dfrac{-\sqrt{3}}{2}\right)\right)$

F has local maximum at $x=\dfrac{5\Pi}{6},\dfrac{7\Pi}{6}$

59. Given, $|-z+2|=1 \qquad m\left(\dfrac{z+1}{z+2}\right)=\dfrac{1}{5}$

Let $z=x+iy$ then $m\left(\dfrac{x+1+iy}{(x+2)+iy}\right)=\dfrac{1}{5}$

$\Rightarrow \dfrac{(x+2)y-(x+1)y}{(x+2)^2+y^2}=\dfrac{1}{5}$

$\Rightarrow 10y-5y=\ (x+2)^2+y^2$

$\Rightarrow (x+2)^2+y^2-5y=0$

$|z+2|=1 \quad \Rightarrow (x+2)^2+y^2 2 \quad \Rightarrow y=\dfrac{1}{5}$

$\Rightarrow (x+2)^2=1-\dfrac{1}{25}=\dfrac{24}{25}$

$\left|\mathrm{Re}\left(\bar{z}+2\right)\right|=\left|\mathrm{Re}(x+2-iy)\right|=|x+2|=\sqrt{\dfrac{24}{25}}=\dfrac{2\sqrt{6}}{5}$

60. $A=\begin{bmatrix}2 & a & 0\\ 1 & 3 & 1\\ 0 & 5 & b\end{bmatrix}$ *Given* $A^3=4A^2-A-21I$

$\Rightarrow A^3-4A^2+A+21I=0$

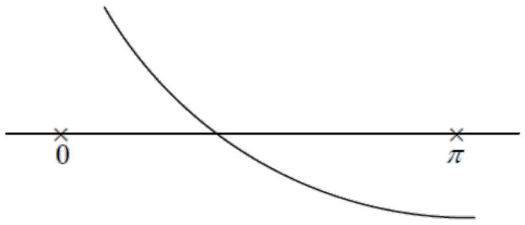
$T\gamma(\mathrm{A})=4\Rightarrow 2+3+\mathrm{b}=4\Rightarrow \mathrm{b}=-1$

$|A|=2(3b-5)-a(\mathrm{b})=-21$

$\Rightarrow -6-10\ -\mathrm{a}(-1)=-21 \quad \Rightarrow a=-5$

$2a+3b=\ 2(-5)+3(-1)=\ -13$

61.



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$f(x)=\cos x-x+1$

$f'(x)=-\sin x-1$

$f(0)=2; f(\pi)=-1$

$\Rightarrow f(0)=0$ exactly once in $[0,\pi]$

$\therefore S_1$ is correct and S_2 is incorrect

62. $$(A)=\int_{-\pi}^{-\frac{3\pi}{4}}\cos x dx-\int_{-\frac{3\pi}{4}}^{0}\sin x dx+\int_{0}^{\frac{\pi}{4}}\sin x dx+\int_{\pi/4}^{\pi/2}\cos x dx-\int_{\pi/2}^{\pi}\cos x dx$$

$$=-[\sin x]_{-\pi}^{-3\pi/4}+[\cos x]_{-3\pi/4}^{0}+[\cos x]_{0}^{\pi/4}+[\sin x]_{\pi/4}^{\pi/2}-[\sin x]_{\pi/2}^{\pi}$$

$$=\frac{1}{\sqrt{2}}+1+\frac{1}{\sqrt{2}}+\left(1-\frac{1}{\sqrt{2}}\right)+\left(1-\frac{1}{\sqrt{2}}\right)+1$$

$A=4 sq uni$

$A^2=16$

63. Equation of hyperbola is

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$$

$$e=\sqrt{\frac{a^y b^2}{b^2}}=\sqrt{3}; \quad \frac{2a^2}{b}=4\sqrt{3}$$

$\Rightarrow a^2=2b^2 \Rightarrow a^2=6;\ b^2=3$

Equation of hyperbola is $\frac{x^2}{6}\frac{-y^2}{3}=-1$

$(\alpha, 6)$ lies on if $\Rightarrow \frac{\alpha^2}{6}\frac{-36}{3}=-1$

$\frac{\alpha^2}{6}=11 \qquad \alpha=\sqrt{66}$

$\beta=(ey_1+b)(ey_1-b)=e^2y_1^2-b^2$

$3(36)-3=105$

$\therefore \alpha^2+\beta=66+105=171$

64. $\bar{a}=\alpha t\hat{i}+6\hat{j}-3\hat{k}$

$\bar{b}=t\hat{i}+2\hat{j}-2\alpha t\hat{h}$

$\bar{a}.\bar{b}<0\ \forall t\in R$

$\Rightarrow \alpha t^2+6\alpha t-12<0 \qquad \forall t\in R$

$\Rightarrow \alpha<0 \qquad 36\alpha^2+48\alpha<0$

$\Rightarrow 12\alpha(3\alpha 4)<0$

$\Rightarrow \alpha\in\left(\frac{-4}{3},0\right) also\ \ \alpha=0$

$\therefore \alpha\in\left(\frac{-4}{3},0\right]$

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65. Let a, b be the numbers

$a+b=24$

$ab \geq \frac{3}{4}(144) \Rightarrow ab \geq 108$

$a(24-a) \geq 108$

$\Rightarrow a^2-24a+108 \leq 0$

$(a-6)(a-18) \leq 0$

$a \in [6,18]$

$req \;\; prob \frac{13}{23}$

66. $mx = \frac{-3}{5} \;\; \pi < x < \frac{3\pi}{2}$

$\Rightarrow \tan x = \frac{3}{4};\; \cos x = \frac{-4}{5}$

$80\left(\frac{9}{16}+\frac{4}{5}\right) = \;\; 45+64=109$

67. $P=(x,y,z) \qquad Q=(x,y,o)$

$OP = \sqrt{x^y y^2+z^2} =$

$\cos\theta = \frac{x}{\sqrt{x^y y^2}} \;;\; \cos\varphi = \frac{z}{\sqrt{x^y y^2+z^2}}$

Distance of P $x\,axis$ is $\sqrt{y^2+z^2}$

$\cos^2\theta \sin^2\phi = \frac{x^2}{x^y y^2}\left(\frac{x^y y^2}{x^y y^2+z^2}\right) = \frac{x^2}{x^y y+z^2}$

$1-\cos^2\theta \;\sin^2\phi = \frac{y^2+z^2}{x^y y^2+z^2}$

$\therefore \gamma\sqrt{1-\cos^2\theta \;\sin^2\phi} = \sqrt{x^y y+z^2}\,\frac{\sqrt{y^y z^2}}{x^y yz^2}$

$\therefore required \;\; dis\tan ce = \gamma\sqrt{1-\cos^2\theta\sin^2\phi}$

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68. $I_n = \int_0^1 \left(1-x^k\right)^n dx$

$= \int_0^1 \left(1-x^k\right)^n 1 \; dx$

$= \left[x\left(1-x^k\right)^n\right]_0^1 - n\int_0^1 \left(1-x^k\right)^n \left(-kx^{k-1}\right)_{dx}^{x}$

$I_n = -nk\int_0^1 \left(1-x^k\right)^{n1}\left(1-x^k-1\right)dx$

$= -nk\int_0^1 \left(1-x^k\right)^n dx + nk\int_0^1 \left(1-x^k\right)^{n-1} dx$

$I_n = -nk\; I_n + nk\; I_{n-1}$

$I_n = \dfrac{nk}{nk+1} I_{n-1}$

$(21k) I_{20} = (21k+1) I_{21} \Rightarrow n = 7$

$(\because 21k = 147)$

69. $\left(1+y^2\right)e^{\tan x}dx + \cos^2 x\left(1+e^{2\tan x}\right)dy = 0$

$\Rightarrow \int \dfrac{\sec^2 x\, e^{\tan x}}{1+\left(e^{\tan x}\right)^2}dx + \int \dfrac{dy}{1+y^2} = const$

$\Rightarrow \tan^{-1}\left(e^{\tan x}\right) + \tan^{-1} y = C$

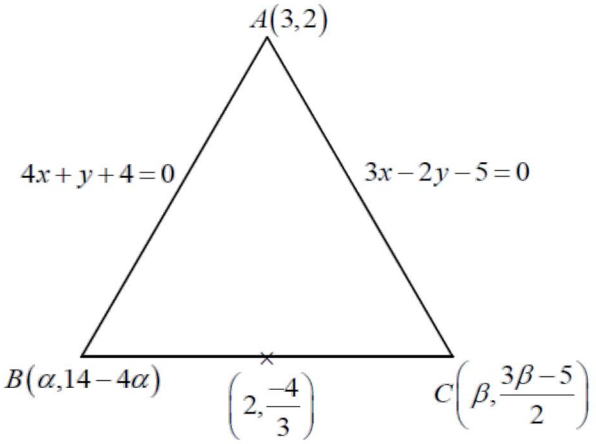
$y(0) = 1 \Rightarrow C = \dfrac{\pi}{4} + \dfrac{\pi}{4} = \dfrac{\pi}{2}$

$\Rightarrow \tan^{-1}\left(e^{\tan x}\right) + \tan^{-1} y = \pi/2$

$\tan^{-1} y(\pi/4) = \dfrac{\pi}{2} - \tan^{-1} e = \cot^{-1} e$

$\Rightarrow y(\pi/4)\tan\left(\cot^{-1} e\right) = \dfrac{1}{e}$

70.



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$$\frac{2\beta+\alpha}{3}=2$$

$$\frac{\frac{2(3\beta-5)}{2}+14-4\alpha}{3}=\frac{-4}{3}$$

$$\left.\begin{aligned}\Rightarrow\alpha+2\beta&=6\\-4\alpha+3\beta&=-13\end{aligned}\right\}\quad\Rightarrow\beta=1;\ \alpha=4$$

$$\therefore B=(4,2)\qquad C=(1,-1)$$

71. $X\to BV$ denoting number of blue balls

$n=3$; $P=$ prob of be the a blue ball $=\frac{5}{9}$

$$\bar{x}=Mean=np=3\left(\frac{5}{9}\right)=\frac{15}{9}$$

$y\to B.V$ denoting number of yellow balls drawn here

$n=3\quad p=$prob of be the a yellow ball $=\frac{4}{9}$

$$\bar{y}=Mean=np=3\left(\frac{4}{9}\right)=\left(\frac{12}{9}\right)$$

$$7\bar{x}+4\bar{y}=7\left(\frac{15}{9}\right)+4\left(\frac{12}{9}\right)=\frac{153}{9}=17$$

72. The last term of K^{st} row

Is $\sum k$ and the 1st term of K^{st} row is $\sum k-(k-1)$

$$\frac{k^2+k}{2}-k+1\le 5310\le\frac{k(k+1)}{2}$$

$$\frac{k^2-k+2}{2}\le 5310\le\frac{k(k+1)}{2}$$

$$\Rightarrow k=103$$

73. $$\alpha=\sum_{r=0}^{n}\left(4r^2+2r+1\right)n_{C_r}$$

$$=4\sum_{r=0}^{n}r^2\left(n_{C_r}\right)+2\sum_{r=0}^{n}r\left(n_{C_r}\right)+\sum_{r=0}^{n}n_{C_r}$$

$$(1+x)^n=n_{C_0}+n_{C_1}x+n_{C_2}x^2\ \square\ +n_{C_n}x^n$$

$$n(1+x)^{n-1}=n_{C_1}+2.n_{C_2}x+.....+n.\ \mathrm{n}_{C_n}x^{n-1}$$

$$put\ x=1\ then\ \sum_{r=0}^{n}rn_{C_r}=n2^{n-1}$$

$$Multiply\ eqn\ 1\ by\ x;\ differentate\ \&\ put\ x=1$$

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$Then \sum_{r=0}^{n} r^2 {}^nC_r = n(n+1)2^{n-2}$

$\therefore \alpha = 4n(n+1)2^{n-2} + 2n\,2^{n-1} + 2^n$

$= \left(n^2 + n + n + 1\right)2^n = (n+1)^2 2^n$

$B = \sum_{r=0}^{n} \frac{{}^nC_r}{r+1} + \frac{1}{n+1}$

$= \frac{1}{n+1} \sum_{r=0}^{n} {}^{n+1}C_r + 1 + \frac{1}{n+1}$

$\Rightarrow \frac{1}{n+1}\left(2^{{}^nC_r} - 1\right) + \frac{1}{n+1} = \frac{2}{n+1}$

$140 < \frac{(n+1)^2 2^{n+1}}{\left(\frac{2n+1}{n+1}\right)} < 281 \;\; 140 < (n+1)^3 < 281 \; n+1 = 6 \Rightarrow n = 5$

74. $L = \lim_{x \to 0} \frac{1 - \cos x (\cos 2x)^{1/2} (\cos 3x)^{1/3} \ldots\ldots (\cos 10x)^{1/10}}{x^2} = \left(\frac{0}{6}\right)$

$= \lim_{x \to 0} \frac{\sin x}{2x} + \frac{\sin 2x}{2x} + \frac{\sin 3x}{3x} + \ldots. + \frac{\sin 10x}{2x}$

$= \frac{1}{2}[1 + 2 + 3 + \ldots + 10] \quad = \frac{1}{2}(55) \qquad \therefore 2L = 55$

75. $f(\theta) = \frac{\sin^4\theta + 3\cos^2\theta}{\sin^4\theta + \cos^2\theta}$

$1 + \frac{2\cos^2\theta}{\sin^4\theta + \cos^2\theta}$

$1 + \frac{2\cos^2\theta}{\left(1 - \cos^2\theta\right)^2 + \cos^2\theta}$

$1 + \frac{2\cos^2\theta}{\cos^4\theta - \cos^2\theta + 1}$

$1 + \frac{2}{\cos^2\theta + \frac{1}{\cos^2\theta} - 1}$

$Min\left(f(\theta)\right) = 1 \;\; When \cos\theta = 20$

$Max\, f(\theta) = 3 \quad When \sin\theta = 0$

$\therefore [\alpha, \beta] = [1,3]$

Since of infinite G.P $64 + \frac{64}{3} + \frac{64}{32} + \ldots. = \frac{64}{\left(1 - \frac{1}{3}\right)} \quad = 96$

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